Aditya Mahadevan
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Education

Stanford University, 2019 – 2025

PhD candidate in physics, advised by Daniel S. Fisher. Research in statistical physics, biophysics, microbial diversity. Thesis title: *Evolutionary Dynamics of Interacting Ecosystems*

Harvard University, 2016 – 2019

AB Physics, 2019 (Advanced Standing)

Schools and Fellowships

Stanford Center for Computational, Evolutionary and Human Genomics (CEHG) graduate fellow, 2024 – 2025
Kavli Institute for Theoretical Physics (KITP) graduate fellow, July – December 2023
KITP quantitative biology summer school, 2021
UMass Amherst summer school on complex fluids and soft solids, 2019

Publications

Mahadevan, A., & Fisher, D. S. (2025). Continual evolution in nonreciprocal ecological models. PRX Life, 3(3), 033008.

Mahadevan, A., Pearce, M. T., & Fisher, D. S. (2023). Spatiotemporal ecological chaos enables gradual evolutionary diversification without niches or tradeoffs. Elife, 12, e82734.

Pandey, A., Mahadevan, A., & Cowsik, A. (2023). Random geometry at an infinite-randomness fixed point. Physical Review B, 108(6), 064201.

Li, F., Mahadevan, A., & Sherlock, G. (2023). An improved algorithm for inferring mutational parameters from bar-seq evolution experiments. BMC genomics, 24(1), 246.

Presentations

Unifying Theories in High-Dimensional Biophysics, ICTS Bangalore, Summer 2025

Poster on continual evolution in ecological models

American Physical Society March Meeting, Las Vegas, Spring 2023

Contributed talks: *High-Diversity Coevolutionary Dynamics in a Model of Interacting Bacteria and Phage*, *Random Geometry at an Infinite-randomness Fixed Point*

Gordon Research Conference on Marine Microbes, Switzerland, Spring 2022

Poster on eco-evolutionary dynamics in host-pathogen systems

American Physical Society March Meeting, Boston, Spring 2019

Contributed talk: *Scattering from Monolayer and Bilayer Step Edges on Topological Sb(111)*

Teaching

TA for Randomness & Renormalization Group (Stanford), Winter 2025
TA for Mechanics *Physics 41* (Stanford), Winter 2023
CA for Cellular Biophysics *Applied Physics 294* (Stanford), Spring 2022
TA for Modern Physics *Physics 26* (Stanford), Spring 2021
TA for Electromagnetism *Physics 43* (Stanford), Spring 2020
CA for Physics of Waves *Physics 15c* (Harvard), Fall 2018
CA for Mechanics Lab *Physics 15a* (Harvard), Spring 2018
TA for Theoretical Computer Science *CS121* (Harvard), Fall 2017 (Derek Bok certificate of distinction)

Outreach and Activities

Co-President of Stanford Science Bus, 2020 – present

Coordinator and teacher for science outreach program in East Palo Alto, planning and leading science experiments for elementary school students.

Mentor for Stanford FAST, 2022 – 2023

Mentored a high school student from San Jose on a physics research project of his own design.

Instructor for Stanford Splash, Summer 2022
Planned and taught a course on dimensional analysis for high school students.

Instructor for Rainstorm Teaching program, Summer 2020
Co-planned and taught a course on chaos for high school students.

Volunteer with Pudiyador, 2020 – 2021
One-on-one mentoring over the phone with an elementary school student in Chennai, teaching English and providing educational support during the extended closure of school due to COVID.

Volunteer with Calico Youth Services, 2020 – 2022
One-on-one mentoring with students from the Bay Area, working on math, English, reading and writing skills.

Volunteer with Mission Hill After School Program, 2017 – 2019
Volunteer with after school program in Mission Hill, Boston, once a week, working with elementary school students on schoolwork and reading.

Music & Languages composition, piano, violin, mridangam, Spanish, Hindi, Tamil